5 QUANTA OF TIME

There's an amphora
of old wine in the house
of nine years' vintage.
There is in the garden, Phyllis,
laurel for braiding crowns
and much ivy . . .
I invite you to celebrate
this day in mid-April—
a festive one for me,
dearer than my own birthday. (IV, 11)

The strange landscape of the physics of relativity that I have described so far becomes even more alien when we consider quanta and the quantum properties of space and of time.

The discipline that studies these is called "quantum gravity," and this is my own field of research.[50] There is not yet a theory of quantum gravity that has been generally accepted by the scientific community, or has obtained experimental support. My scientific life has been largely dedicated to contributing to the construction of a possible solution to the problem: *loop* quantum gravity, or *loop theory*. Not everyone is betting on this turning out to be the right solution. Friends who work on string theory, for instance, are following different paths, and the battle to establish who is right is still raging. This is good—science also grows thanks to fierce debates: sooner or later, it will become clear which theory is correct, and perhaps it will not be long.

Divergences of opinion regarding the nature of time, however, have diminished in the last few years, and many conclusions have become reasonably clear to most. What has been clarified is that the residual temporal scaffolding of general relativity, illustrated in the previous chapter, also falls away if we take quanta into account.

Universal time has shattered into a myriad of proper times, but if we factor in the quanta, we must accept the idea that each of these times, in turn, "fluctuates" and is dispersed as in a cloud—and can have only certain values and not others. . . . They can no longer manage to form the spatiotemporal sheet outlined in the previous chapters.